



ENGINEER & MARVEX IDEAS COMPETITION: PITCH FOR TOMORROW

THEME: The Responsive Skin
FOCUS: Passive Cooling via Kinetic Design

TOPIC:

**SMART SENSOR FUSION & SOLAR TRACKING FOCUSING ON THE
“MECHATRONIC” BRAIN**

Number of subtopic: ____2____

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1. Executive Summary

With the Energy Efficiency and Conservation Act (EECA) 2024 mandating strict Building Energy Index (BEI) targets, adaptive facades in tropical climates remain a significant liability. Current systems fail because they rely on temperate-zone assumptions, fragile single-objective control loops and blind trust in ideal sensor conditions, leading to incorrect decisions and unnecessary mechanical wear. To address this, we propose “**The Neuro-Skin Array**,” an anticipatory, decentralised decision system for tropical high-rises built around three key innovations: a **Three-Tier Decision Brain** that integrates sensor distrust, multi-objective optimisation and mechanical restraint; a **Decentralised 4×4 Facade Array** that enables each facade zone to respond independently to local environmental conditions; and an **Interactive Digital Twin** that visualises real-time zone-level conditions, optimisation decisions and facade responses for transparent, explainable control.

Recognizing that historical kinetic facades die from mechanical fatigue rather than optical failure, the Neuro-Skin Array replaces competing control loops with a unified, normalized multi-objective cost function:

$$J(\alpha) = w_T \cdot Q_{\text{pred}}(\alpha) + w_L \cdot E_{\text{error}}(\alpha) + w_M \cdot |\alpha - \alpha_{\text{current}}| + w_A \cdot A_{\text{risk}}(\alpha)$$

This allows the central brain to simultaneously co-optimize shading, daylighting, mechanical wear and safety as a single, mathematically cohesive decision. Validated through a software digital twin, the system shifts the engineering focus away from physical kinetic gimmicks and toward verifiable, software-driven energy reduction. By delivering intelligent peak cooling load shaving and drastically lowering maintenance overhead, the Neuro-Skin Array provides a robust, fault-tolerant pathway for buildings to achieve mandatory EECA compliance.

2. Problem Statement

2.0 Problem Statement:

Static building facades in tropical climates can contribute significantly to solar heat gain, increasing HVAC demand and reducing building energy efficiency. Our challenge is to design a modular, sensor-driven kinetic facade system for the ST Diamond Building in Putrajaya that autonomously adapts its shading configuration to real-time solar conditions, while balancing thermal performance, daylight, mechanical movement and safety.

2.1 Introduction

The enactment of the Energy Efficiency and Conservation Act (EECA) 2024 has shifted building energy performance in Malaysia from a voluntary aspiration to a regulatory requirement. For landmark commercial buildings such as the **ST Diamond Building** in Putrajaya, its distinctive faceted facade presents both an opportunity and a challenge: while its geometry can support adaptive solar-shading strategies, large exposed surfaces remain susceptible to solar heat gain, increasing cooling demand and affecting occupant comfort. Adaptive kinetic facades therefore offer a promising pathway toward improving building energy performance, but their effectiveness depends heavily on the intelligence controlling them.

Historically, kinetic facades have faced operational challenges that can **turn energy-saving systems into expensive, maintenance-intensive assets**. The fundamental issue is not necessarily the mechanical panels themselves, but the naive mechatronic intelligence controlling them. Conventional adaptive systems often rely on fragmented, single-objective control loops, assume ideal sensor conditions and continuously chase small optimisation gains. In a tropical environment such as Putrajaya, these assumptions can lead to unnecessary actuation, sensor-driven errors and excessive mechanical wear, ultimately compromising both energy performance and system reliability.

For the ST Diamond Building, these challenges are particularly relevant because solar conditions around the facade vary continuously throughout the day. Environmental sensors can drift, degrade or become affected by dust and weather, meaning that a faulty pyranometer could incorrectly interpret low irradiance and command the facade to open during periods of high solar exposure. At the same time, continuously adjusting individual facade elements for marginal improvements can create excessive actuator movement and maintenance demands. Furthermore, tropical conditions differ significantly from the temperate environments for which many conventional facade control strategies were developed. High cloud cover and diffuse radiation, glare management and significant latent cooling loads require a control strategy designed specifically around tropical building conditions rather than simple direct-solar tracking.



Figure 1: ST Diamond Building, Putrajaya, a tropical commercial building whose distinctive faceted facade provides the physical basis for the proposed decentralised kinetic facade system.

Addressing this challenge, our proposal focuses on the **decision brain controlling the facade** rather than the kinetic panels alone. The physical facade acts as the mechatronic substrate, while the proposed Neuro-Skin Array provides the intelligence required to make it resilient. Through a **Three-Tier Decision Brain (Tier 1: Almanac Lie-Detector, Tier 2: Co-Optimisation Engine, and Tier 3: Movement-Budget Safety Gate)**, the system distrusts unreliable inputs, balances competing thermal and daylight objectives, and restrains unnecessary mechanical movement. Combined with a **Decentralised 4×4 Facade Array** and an **Interactive Digital Twin**, the proposed system aims to transform the ST Diamond Building's facade from a **passive solar-exposure surface into an adaptive, energy-efficient and mechanically resilient building system**.

2.2 Objective

The primary objective of the Neuro-Skin Array is to engineer a fault-tolerant, anticipatory decision brain for the ST Diamond Building, Putrajaya, unifying shading, daylighting and thermal management into a single multi-objective control framework that supports EECA 2024 Building Energy Index (BEI) targets. Rather than treating energy efficiency in isolation, the solution is designed to create sustained value across three pillars: economic, operational efficiency and occupant well-being. Leveraging the building's distinctive faceted geometry as a decentralised 4×4 facade array, the system transforms the facade into an adaptive building skin responsive to tropical environmental conditions.

Within the economic pillar, the framework reduces cooling demand through predictive co-optimisation of solar exposure, thermal load and daylight conditions. By preventing unnecessary facade movements through the Movement-Budget Safety Gate, the system also reduces actuator wear and associated maintenance costs, ensuring that energy savings are not offset by excessive mechanical operation. This enables the ST Diamond Building to achieve more efficient operation while improving the long-term economic viability of its adaptive

facade.

Under the operational efficiency pillar, the framework empowers facility management through automated diagnostics and decentralised resilience. The Almanac Lie-Detector cross-validates pyranometer readings against astronomical solar position to identify unreliable sensors before they trigger incorrect facade movements. Meanwhile, the 16-zone decentralised architecture allows individual sections of the ST Diamond Building's facade to continue operating independently during communication disruptions, reducing reliance on a single point of failure and improving system resilience.

For occupant well-being, the Co-Optimisation Engine balances daylight, glare and thermal conditions across individual facade zones rather than optimising a single parameter in isolation. Deliberate and controlled facade movements minimise unnecessary visual and acoustic disturbance while maintaining a more stable indoor environment. Supported by the Interactive Digital Twin, facility managers can also visualise zone-level environmental conditions and facade responses in real time, making the system's decisions transparent and easier to monitor.

3. Research & Literature Review

In Southeast Asian tropical climates, the building sector accounts for approximately 30% of primary energy demand, driven predominantly by the necessity for mechanical cooling. (*Energy System - IEA*, n.d.) In hot-humid environments like Malaysia and Indonesia, energy-intensive vapour-compression HVAC systems are responsible for 50% to over 60% of a commercial building's total electricity consumption. (Ameen, 2005) The persistent high ambient temperatures and dominant latent cooling loads created by high humidity pose a perpetual operational burden on high-rise structures. As urbanization accelerates and mandatory legislative frameworks, such as Malaysia's Energy Efficiency and Conservation Act (EECA) 2024, come into enforcement, mitigating envelope heat gain is no longer optional. (*MySTEE : Info*, n.d.) While basic architectural adjustments offer some relief, they are increasingly insufficient to meet aggressive new Building Energy Index (BEI) targets, necessitating advanced mechatronic interventions at the building envelope.

Before introducing active kinetic systems, the efficacy of passive cooling and daylighting balance must be established. Passive strategies in tropical architecture traditionally rely on fixed horizontal shading, optimized building orientation to minimize east-west exposure and calculated window-to-wall ratios to mitigate sensible heat gain while permitting diffuse natural light. The ST Diamond Building in Putrajaya stands as a regional benchmark for this approach. Its inverted diamond geometry, split-window design, insulated roof and daylighting strategies reduce solar heat gain while supporting natural illumination. In addition, its radiant slab cooling system, using chilled-water pipes embedded within the concrete floor slabs, leverages thermal mass to provide efficient cooling and shift cooling demand. These integrated strategies contribute to the building's reported operational BEI of approximately 65 kWh/m²/year, significantly below the cited Malaysian office benchmark of 210 kWh/m²/year. (Lim & Lim, 2016)

However, the ST Diamond Building's facade is fundamentally static. Its geometry provides permanent self-shading but cannot dynamically respond to changing cloud cover, solar intensity, glare or occupancy conditions. This creates an opportunity for an adaptive layer that complements rather than replaces its existing passive design and radiant cooling strategy.

To surpass the limitations of static architecture, global projects have deployed kinetic facades that adapt to environmental conditions in real time. Systems such as the Al Bahar Towers demonstrate that dynamic shading can reduce solar heat gain and cooling energy consumption by upwards of 20%. (*Al Bahar*, 2022) However, many kinetic systems rely on continuous reactive control loops that hunt for micro-optimizations, leading to excessive movement, actuator fatigue and high maintenance costs. (Stiefelmaier et al., 2024) For the ST Diamond

Building, simply adding kinetic panels could therefore undermine its existing efficiency if movements are not carefully controlled.

The fundamental gap in current adaptive facade technology is the reliance on control logic ported from temperate climates, which falsely assumes ideal sensor conditions and prioritizes direct solar tracking. (Lee et al., 2021) At near-equatorial latitudes like Malaysia ($\sim 3^{\circ}\text{N}$), direct solar tracking is a relatively straightforward geometric calculation, while diffuse radiation, high humidity and latent cooling loads require more adaptive decision-making. Existing systems also use fragmented, single-objective control loops where thermal and daylighting objectives can conflict, causing unnecessary facade movement. (Alsukkar et al., 2025) Therefore, a critical need exists for a decentralised mechatronic decision brain that unifies shading, daylighting, thermal performance and mechanical wear into a single real-time optimisation framework.

The proposed approach also complements the ST Diamond Building's existing integrated-design philosophy. By reducing solar heat gain at the facade, the system can work alongside the building's radiant slab cooling, while avoiding excessive shading that could increase artificial lighting demand. The proposed multi-objective framework therefore treats thermal performance, daylight and mechanical movement as interconnected variables rather than isolated objectives.

Recent work confirms that multi-objective optimization is an appropriate approach for facade design, jointly balancing energy, daylight and comfort (Le et al., 2022; Nazari et al., 2023; Wagiri et al., 2024). However, these studies largely focus on offline design optimization and assume reliable sensor data, while actuator fatigue remains insufficiently addressed (Stiefelmaier et al., 2024; Formentini & Lenci, 2017). This proposal extends the literature by bringing multi-objective optimisation into real-time control for the ST Diamond Building, treating mechanical wear as an additional optimisation objective and cross-validating physical sensors against astronomical ephemeris data. Thus, the Neuro-Skin Array aims to add an intelligent adaptive layer to the ST Diamond Building while preserving the strengths of its existing passive facade and radiant cooling system.

4. Proposed Solutions

4.1 Concept Overview

To overcome the operational and mechanical limitations of traditional kinetic facades, this proposal introduces “The Neuro-Skin Array,” a decentralised, anticipatory decision brain designed for the ST Diamond Building, Putrajaya, leveraging its distinctive geometry as a 16-zone (4×4) adaptive facade array. The system integrates three decision tiers: an Almanac Lie-Detector that cross-validates sensor data against astronomical predictions, a Co-Optimisation Engine that balances thermal load, daylight, movement and risk, and a Movement-Budget Safety Gate that suppresses unnecessary actuation and enforces deterministic fail-safe behaviour during emergencies. The system operates across independent facade zones, while an Interactive Digital Twin visualises real-time conditions and responses, transforming the ST Diamond Building into an intelligent, resilient and energy-efficient adaptive facade.

4.2 Working Principles

The Neuro-Skin Array operates as a continuous, decentralised control system across the ST Diamond Building, Putrajaya, using its distinctive faceted facade as a 16-zone (4×4) modular kinetic array. At each control cycle, local edge nodes collect environmental data and process each zone through a layered decision architecture comprising the Application Layer, represented by the Interactive Digital Twin for real-time visualisation and self-logging; the **Brain**, which performs multi-objective co-optimisation of thermal load, daylight and mechanical considerations; the Trust Layer, which validates sensor inputs and enforces safety constraints; and the **Body**, consisting of local mechatronic actuators that physically adjust each facade zone. Distributed edge controllers, environmental sensors and motor-current monitoring allow individual zones to respond independently while remaining coordinated with the building-level objective. The modular architecture also allows sensors or actuators to be serviced or replaced without disrupting the wider facade network, ensuring the ST Diamond Building can continue adapting reliably even when individual zones experience hardware or communication failures.

4.3 Feasibility, Safety and Compliance Considerations

The technical feasibility of the Neuro-Skin Array is anchored in its software-first approach, shifting the engineering complexity from bespoke mechanical systems to validated computational logic designed for the ST Diamond Building, Putrajaya. Feasibility is supported through Python-based simulations of representative tropical conditions, evaluating cooling load, daylight performance and actuator movement against conventional control strategies. The proposed 16-zone (4×4) architecture uses lightweight edge-computing controllers, standard communication protocols and off-the-shelf mechatronic components, making the system modular, scalable and practical to implement across the building’s distinctive facade. Safety is embedded through a deterministic priority hierarchy in which safety constraints always override optimisation; during high-wind events, communication failures or power loss, local zones can transition to a predefined fail-safe geometry through unpowered spring-return or gravity-assisted mechanisms. Combined with sensor validation and motor-current monitoring for early fault detection, this decentralised architecture improves reliability while reducing unnecessary

maintenance and supporting the long-term operational viability of an adaptive facade for the ST Diamond Building.

4.4 Unique Value Proposition

The unique value proposition of the Neuro-Skin Array lies in its engineering philosophy of distrust, optimisation and restraint, specifically applied to the distinctive facade of the ST Diamond Building, Putrajaya. Rather than relying on reactive controllers that continuously chase small changes, the system combines a Three-Tier Decision Brain with a decentralised 16-zone (4×4) facade array to make deliberate, locally informed decisions. The Almanac Lie-Detector provides analytical redundancy by cross-validating physical sensor readings against astronomical solar predictions, while the Co-Optimisation Engine balances thermal performance, daylight and mechanical movement within a single decision framework. The Movement-Budget Safety Gate further prevents unnecessary actuation, reducing mechanical wear and improving long-term reliability. Supported by an Interactive Digital Twin, the system makes these decisions visible and explainable, transforming the ST Diamond Building's distinctive facade from a passive architectural feature into a resilient, intelligent and self-preserving building skin designed for the realities of tropical operation.

5. Social, Economy and Environmental Impact Analysis

5.1 Social, Industrial and Community Impact

The Neuro-Skin Array is modelled on the ST Diamond Building in Putrajaya, the Energy Commission’s own dual-Platinum headquarters and the regional benchmark for tropical energy efficiency. Choosing a public, occupied flagship as the design baseline foregrounds the social dimension. The system continuously maintains a visual-comfort daylighting target above 500 lux while suppressing direct glare and localised radiant thermal spikes, protecting the well-being and cognitive performance of the civil servants who occupy the building. From an industrial perspective, a software-first decentralised digital-twin architecture shifts facilities management from reactive manual maintenance toward predictive data analysis and digital-twin supervision. Besides, since the demonstrator is a government building, it sets a visible replication example for the wider public-sector estate. Finally, equity and public safety are structurally protected. The deterministic safety override holds the envelope in a fail-shaded geometry during severe tropical weather, protecting occupants inside and the pedestrian community below from uncontrolled panel motion.

Beyond skills, the system creates jobs across its value chain, including facade installation and commissioning crews during rollout, a standing predictive-maintenance and building-analytics function during operation, and upstream demand for a domestic supply chain in recycled-aluminium louvres and edge-electronics assembly. Table 5.1 below summarises the human-centred performance metrics the design is held to.

Table 5.1 Human-centred performance metrics

Metric	Target/Performance Mechanism
Daylight quality	Maintain ≥ 500 lux at the work plane while suppressing direct-beam glare through macro-angle control, complementing ST Diamond’s existing atrium daylighting.
Thermal comfort	Eliminate localised radiant spikes and hold envelope solar gain within budget at peak irradiance, stabilising temperatures near the inclined glazed facade.
Occupant & public safety	Deterministic fail-shaded geometry on high wind, sensor fault or power loss via gravity/spring return without uncontrolled panel motion above pedestrians.

5.2 Economic and Financial Assessment

The economic model is built directly on the ST Diamond Building’s published figures, which is a deliberate rigour choice. Rather than a hypothetical tower, this baseline is a real, instrumented, fully-documented building. ST Diamond has a gross floor area of 14,230 m² and an actual Building Energy Index of 65 kWh/m²/year (Suruhanjaya Tenaga, n.d.), roughly a third of the 210 kWh/m²/year typical of a Malaysian office. Total electricity is therefore about 925 MWh/year, of which mechanical cooling accounts for roughly half, or about 462 MWh/year; at a blended commercial tariff of RM 0.40/kWh (TNB, 2025) that is a baseline cooling cost of about RM 185,000/year. Critically, ST Diamond is already best-in-class. Its 25° self-shading tilted facade and night-cooled radiant slab do most of the passive work so any facade saving here is strictly incremental over an excellent static baseline, and is presented as such.

The facade's contribution is reported strictly as thermal load shaved and shifted, converted to electricity linearly through the chiller coefficient of performance ($COP \approx 3.5$). We deliberately do not amplify a cooling-load reduction into a larger electrical saving using fan- or pump-affinity (cube) relationships. At the plant scale, chiller energy tracks the removed cooling load far more linearly than the cubic part-load curve of a single fan or pump, so applying the affinity law here would materially overstate the benefit. On a building that already night-shifts 40% of its cooling into the concrete slab, this restraint matters more, not less.

On this basis, we model a conservative 6% incremental reduction in annual cooling energy, which is lower than a naive glazed tower precisely because ST Diamond's static shading is already good, capturing the transient cloud-cover, diffuse-sky and dynamic-glare response in which a fixed 25° tilt cannot. That is approximately equal to 27,700 kWh/year ($\approx$ RM 11,100/year), plus a peak-demand shave of approximately RM 6,500/year and modest additional off-peak arbitrage of approximately RM 2,500/year, which is about RM 20,000/year in energy-related saving. The decisive lever, however, is independent of the building's efficiency. A reactive kinetic facade of this size incurs on the order of RM 150,000/year in actuator replacement, which the movement-budget gate cuts to $\approx$ RM 50,000/year, saving $\approx$ RM 100,000/year. This is why the intelligence layer, not the panels, carries the case.

Annual figure (ST Diamond, 14,230 m ²)	Static facade (baseline)	Reactive kinetic facade	Neuro-Skin Array
Cooling electricity cost	RM 185,000	RM 174,000*	RM 165,000
Facade maintenance	Approx. RM 0 (no actuators)	RM 150,000 (burnout)	RM 50,000 (budgeted wear)
Realised energy saving	Baseline (best-in-class)	Intended, often lost to deactivation	RM 20,000 (reliably realised)
Intelligence-layer CapEx (vs reactive)	-	-	+ RM 300,000 (one-off)
Payback on the intelligence layer	-	-	$\approx$ 2.5 years

**Intended saving before deactivation. Figures are engineering estimates from the model in Appendix A; energy savings use a linear COP conversion (no affinity-law amplification).*

The innovation therefore returns its investment in about two and a half years, with an incremental RM 300,000 in sensors, edge nodes and software against $\approx$ RM 120,000/year in combined maintenance-avoidance and smarter-control savings, giving a 15-year net of roughly +RM 1.5 million on the intelligence layer alone. **The same brain can be licensed as a Facade-as-a-Service (FaaS) plug-in to existing Building Management Systems, separating high-margin software from physical substrate.** It must be said plainly that energy savings on a building as efficient as ST Diamond are modest by design; the value here is lifecycle survival plus demonstrator credibility on the Energy Commission's own flagship.

For scalability, this is the point ST Diamond proves. If the system pays even at the passive ceiling of 65 kWh/m²/year, the far larger stock of typical 210 kWh/m²/year towers, with roughly 3.2 $\times$ the cooling energy and abundant unshaded glazing, benefits proportionally more, on the

order of RM 36,000/year of energy saving on a similarly-sized building, before the same maintenance and demand benefits are counted. Demonstrating value at the benchmark is the strongest possible argument for the mass-market retrofit.

For material sourcing, the kinetic louvres are the largest material cost and embodied-carbon item, so the sourcing choice is assessed explicitly below. Specifying ~80% recycled aluminium extrusion rather than primary metal is broadly cost-neutral. Secondary aluminium trades at a modest discount to primary ingot on the LME, while cutting embodied carbon by roughly 70%, because remelting scrap needs only about 5% of the energy of primary electrolysis (International Aluminium Institute, 2024).

Louvre material basis	Conventional (primary Al extrusion)	Recycled Al extrusion (~80%)
Indicative material cost	≈ RM 12–13 / kg	≈ RM 11–12 / kg (secondary discount)
Embodied carbon (cradle-to-gate)	≈ 10.5 kgCO ₂ e / kg	≈ 3 kgCO ₂ e / kg
Production energy	Baseline (primary electrolysis)	≈ 5% of primary (remelt)
Net effect	-	≈ 70% lower embodied carbon at comparable or lower cost

5.3 Sustainability & Environmental Impact

The sustainability case has 2 distinct components, and separating them avoids the common overclaim. The operational component is a direct Scope 2 reduction. The modelled 6% cut in ST Diamond’s cooling energy (≈ 27,700 kWh/year) avoids roughly 20 tCO₂ per year, using the Peninsular Malaysia grid emission factor of 0.740 kgCO₂/kWh (Energy Commission, 2024), which is a modest figure, honestly, because the building is already efficient. The distinctive component is embodied carbon and e-waste. By enforcing a movement budget, the system prevents the premature destruction of hundreds of motors, microcontrollers and wiring harnesses; extending actuator life roughly threefold avoids on the order of two full mechatronic replacement cycles across a ~250-node fleet over 15 years, a benefit reactive systems cannot claim regardless of optical performance.

2 further dimensions complete the picture. On material embodied carbon, sourcing the louvres in recycled rather than primary aluminium is a significant one-off saving. For a representative, ~4,000 m² louvre array of roughly 32 tonnes, the ~7.5 kgCO₂e/kg reduction shown in Section 5.2 avoids on the order of 240 tCO₂e of embodied carbon at construction. On renewable resource usage, the system’s modest edge-node and actuator draw can be supplied directly from ST Diamond’s existing 71.4 kWp rooftop photovoltaic array. More importantly, by shaving and shifting the cooling peak it improves the effective utilisation of on-site renewables and reduces the building’s call on fossil-heavy peaking generation, while its recycled-aluminium and disassembly-for-reuse design keeps the material in a circular loop rather than drawing on virgin ore.

Sustainability Category	Estimated Impact & Strategy	SDG Alignment
Operational Carbon Footprint	$\approx 20 \text{ tCO}_2/\text{yr}$ avoided (6% of 462 MWh/yr cooling $\times 0.740 \text{ kgCO}_2/\text{kWh}$, EC 2024). Reported as load shaved/shifted, not affinity-amplified.	SDG 7: Affordable and Clean Energy
Embodied carbon / e-waste (LCA)	$\sim 3\times$ actuator life via movement-budget gating avoids ≈ 2 replacement cycles of a ~ 250 -node fleet over 15 yr; recycled louvres avoid $\approx 240 \text{ tCO}_2\text{e}$ embodied at construction.	SDG 12: Responsible Consumption
Material Sourcing	Low-draw electronics run off ST Diamond's 71.4 kWp PV; louvres in $\sim 80\%$ recycled aluminium, designed for non-destructive disassembly and end-of-life recycling.	SDG 9: Industry, Innovation & Infrastructure
Urban Resilience	Decentralised intelligence prevents total envelope failure; dynamic solar reflectance mitigates local urban-heat-island contribution.	SDG 11: Sustainable Cities & Communities

6. Implementation Strategy & Feasibility

6.1 Business Justification

Anchoring the proposal to the ST Diamond Building sharpens rather than weakens the business case because it separates two value streams cleanly. The energy saving on a best-in-class building is modest but the maintenance saving is not. The movement-budget gate cuts actuator upkeep by roughly RM 100,000/year against a reactive baseline, which drives the ≈ 2.5 -year payback on the intelligence layer. The strategic value is threefold. Firstly, for demonstrator credibility, proving the system on the Energy Commission’s own dual-Platinum flagship is the strongest possible reference for the wider retrofit market. Secondly, regulatory positioning under EECA 2024, where the qualifying green capital expenditure lets the owner claim the Green Investment Tax Allowance (GITA), which is a 100% allowance on qualifying CapEx offsettable against up to 70% of statutory income (MGTC, 2024). Thirdly, for the scalability, the same brain applied to the large stock of typical 210 kWh/m²/year towers delivers several times the energy saving shown here, turning a modest benchmark result into a substantial mass-market opportunity.

6.2 Project Phases and Planning

The deployment roadmap follows a four-phase, software-first execution strategy over 18 months, validating the intelligence against ST Diamond’s published performance data before committing to hardware. Phase 1 (Months 1–3) is dedicated to algorithmic R&D and software validation, finalising the multi-objective cost function and proving the Almanac Lie-Detector against synthetic tropical-weather data benchmarked to the ST Diamond baseline. Phase 2 (Months 4–6) transitions to Hardware-in-the-Loop (HIL) testing on a scaled 2×2 (four-zone) prototype to validate motor-current sensing and the deterministic safety overrides. Phase 3 (Months 7–12) executes a targeted pilot on a single high-exposure elevation of a commercial partner, measured against the ST Diamond reference to establish a real-world digital-twin baseline. Phase 4 (Months 13–18) delivers full 16-zone commercial integration and scaling, using data analysts, facade installers and BMS-integration specialists.

Phase / Activity	M1–3	M4–6	M7–9	M10–12	M13–15	M16–18
Phase 1: Algorithm R&D & software validation						
Phase 2: Hardware-in-the-Loop (2×2 prototype)						
Phase 3: Pilot deployment & digital-twin baseline						
Phase 4: Full 16-zone integration & scaling						

Figure 6.1: 18-month deployment Gantt (shaded cells indicate active periods).

Each phase carries distinct partner and manpower requirements, with the pilot in Phase 3 acting as the go/no-go gate before full commercial scaling.

Phase	Key partners / stakeholders	Resources & manpower
P1 (M1-3)	University research lab; Energy Commission (ST Diamond benchmark data); weather-data provider.	2 control-software engineers, 1 data scientist; simulation compute and ephemeris datasets.
P2 (M4-6)	Precision-manufacturing partner; mechatronics / motor supplier.	1 mechatronics engineer, 1 test engineer; 2×2 HIL rig, motor-current sensors, edge controllers.
P3 (M7-12)	Commercial building owner (pilot host); BMS vendor; facade contractor.	Installation crew, 1 BMS integrator, 1 data analyst; single-elevation pilot hardware and metering.
P4 (M13-18)	Developer / asset owner; MGTC (GITA); facade installers; recycled-aluminium supplier.	Facade installers, BMS specialists, 1–2 data analysts; full 16-zone hardware fleet.

6.3 Risk Identification and Mitigation Strategies

To guarantee technical feasibility, commercial viability and operational manageability, the project proactively addresses failure modes across all deployment domains. The engineering philosophy of distrust and restraint is the primary defence against the technical and performance risks that typically plague kinetic architecture, ensuring the facade fails safely rather than failing open.

Risk category	Identified risk	Mitigation strategy
Technical	Physical sensor drift or fouling leading to disastrous, unintended panel actuations.	The Lie-Detector: cross-validate pyranometer data against an immutable astronomical ephemeris to instantly filter faulty inputs.
Performance	Machine-learning prediction errors causing the system to over-shade or under-shade.	Deterministic overrides: bound ML outputs strictly within physics-based safety constraints; a bad prediction can never override safety.
Financial	Initial CapEx overruns from the high cost of bespoke kinetic components.	Modular sourcing: off-the-shelf industrial edge microcontrollers and recycled modular aluminium, supported by GITA tax incentives.
Schedule	Hardware supply-chain delays stalling the building-integration timeline.	Software-first phasing: advance the digital twin and control logic independently of hardware delivery for immediate deployment on arrival.
Market	Marginal energy gains on already-efficient buildings like ST Diamond undermining the retrofit pitch.	Baseline-and-scale: use ST Diamond as the credibility benchmark and target the large stock of typical 210-BEI towers, where cooling headroom and therefore savings are several times larger.

7. Conclusion

The ST Diamond Building represents the ceiling of fixed passive design in the tropics, which is a dual-Platinum benchmark whose 25° self-shading facade and night-cooled radiant slab already achieve a Building Energy Index of 65 kWh/m²/year. Using it as the design baseline is a deliberate test. If a dynamic control brain can add measurable value even here, the case for the far larger stock of poorly-performing towers is overwhelming. The Neuro-Skin Array is engineered for exactly that residual, where the transient cloud-cover, diffuse-sky and dynamic-glare response a static tilt cannot capture, through an Almanac Lie-Detector that ignores degraded sensors, a Movement-Budget Gate that rations mechanical wear, and a unified multi-objective algorithm that balances thermal, daylight, movement and safety in a single decision, directly mirroring the objectives set out in the problem statement.

The benefits are claimed with discipline. On ST Diamond, the energy saving is modest by design of about RM 20,000/year, reported strictly as cooling load shaved and shifted with no affinity-law amplification, avoiding roughly 20 tCO₂/year but the same system on a typical 210-BEI tower delivers several times that. The economics rest on lifecycle survival. Roughly RM 120,000/year in avoided maintenance and smarter control that returns the intelligence-layer investment in about two and a half years, while uniquely among kinetic systems, the movement budget also cuts embodied carbon and e-waste. These advantages carry honest trade-offs of higher integration complexity, reliance on edge-computing availability, and minor structural allowances for the distributed nodes.

By leveraging a software-first architecture, modular off-the-shelf components and recycled materials, the Neuro-Skin Array is feasible, scalable and de-risked which is proven, in concept, against a building that already demonstrates dynamic shading at its atrium and load-shifting in its slab. In showing that true building performance lies in the intelligence of the decision logic rather than the movement of the panels, and in proving it at the regional benchmark, it offers a financially defensible, failsafe pathway for the wider building stock to meet its EECA 2024 obligations.

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9. Appendices

Appendix A: Techno-economic & Carbon Model (assumptions and calculation basis)

All financial and carbon figures in Section 5 and Section 6 derive from the transparent model below. The intent is that any judge can reproduce every headline number from these inputs. HVAC benefits are reported as cooling load shaved and shifted, converted to electricity linearly through the chiller COP; no fan- or pump-affinity (cube) amplification is applied.

Parameter	Value	Source / basis
Baseline building	ST Diamond, Putrajaya	Suruhanjaya Tenaga (n.d.)
Gross floor area	14,230 m ²	GBI project data
Baseline BEI (actual)	65 kWh/m ² /yr	Suruhanjaya Tenaga (n.d.)
Baseline total electricity	≈ 925 MWh/yr	BEI × GFA
Cooling share of electricity	50%	Ameen (2005): 50–60%
Baseline cooling electricity	≈ 462 MWh/yr	Derived
Blended commercial tariff	RM 0.40/kWh	TNB RP4 (2025)
Baseline cooling cost	≈ RM 185,000/yr	Derived (ceiling on savings)
Grid emission factor (Peninsular)	0.740 kgCO ₂ /kWh	Energy Commission (2024)
Chiller COP (linear conversion)	3.5	Typical water-cooled plant
Incremental cooling reduction	6% (≈ 27,700 kWh/yr)	Conservative vs best-in-class static shading
→ Energy cost saving	≈ RM 11,100/yr	27,700 kWh × RM 0.40
→ Operational carbon avoided	≈ 20 tCO ₂ /yr	27,700 kWh × 0.740
Peak demand + arbitrage	≈ RM 9,000/yr	Demand shave + off-peak shift
Maintenance saving vs reactive	≈ RM 100,000/yr	3× actuator life (movement budget)
Intelligence-layer CapEx (incremental)	RM 300,000	Sensors + edge nodes + software
Payback on intelligence layer	≈ 2.5 years	RM 300k ÷ ≈ RM 120k/yr benefit

Appendix B: Simulation Results

A.1 Reference scenario results

The four reference scenarios use the same seed-42 synthetic scattered-cloud day, west façade, 25° outward tilt, occupancy scale, wind override, and controller weights.

Scenario	NeuroSkin daylight (%)	Naive daylight (%)	NeuroSkin load	Naive load	NeuroSkin moves	Naive moves	Rejected ticks	SAFE ticks
Overview	100.0	16.7	0.550	0.456	0	2	0	0
Sensor Trust	100.0	51.9	0.544	0.489	0	6	10	0
Optimisation	100.0	16.7	0.550	0.456	0	2	0	0
Safety	46.7	16.7	0.460	0.456	1	2	0	4

NeuroSkin generally preserved more useful daylight and moved less than the naive irradiance-threshold controller. The naive strategy sometimes achieved a lower relative load by closing more aggressively. This is a comfort-load trade-off; the results do not show one controller winning every metric simultaneously.

A.2 Upright-façade co-optimisation result

The upright west-façade counterfactual represents a conventional vertical façade where adaptive control has more work to do.

Metric	NeuroSkin	Naive baseline	Preferred direction
Daylight compliance	70.8%	12.5%	Higher
Mean relative cooling-load index	0.507	0.469	Lower
Full-day movement count	1	2	Lower

Daylight compliance is the percentage of eligible occupied ticks within the **300–700 lux** band. The naive controller's lower load is associated with substantially lower daylight compliance.

A.3 Ten-seed sensitivity check

Ten deterministic synthetic skies were tested on the upright west façade. NeuroSkin matched or exceeded naive daylight compliance in **10 of 10 seeds** and strictly exceeded it in **7 of 10**. NeuroSkin movements were never higher. Its relative load was higher in nine seeds and equal in one; the largest NeuroSkin-to-naive load ratio was 1.128.

Seed	NeuroSkin daylight (%)	Naive daylight (%)	NeuroSkin load	Naive load	Load ratio	NeuroSkin moves	Naive moves
0	94.7	57.9	0.573	0.522	1.098	0	4
1	100.0	55.2	0.557	0.506	1.101	0	4
2	46.7	46.7	0.565	0.552	1.024	0	2
3	57.7	15.4	0.505	0.468	1.079	1	2
4	100.0	40.0	0.549	0.524	1.048	0	2
5	100.0	78.1	0.564	0.540	1.044	0	2
6	100.0	61.1	0.557	0.494	1.128	0	2
7	100.0	66.7	0.555	0.537	1.034	0	2
8	100.0	100.0	0.550	0.550	1.000	0	0
9	68.8	68.8	0.574	0.570	1.007	0	2

Absolute daylight compliance ranged from 46.7% to 100%, confirming that the generated cloud realization materially affects the result. These ten seeds are a bounded sensitivity check, not a statistical confidence interval.

A.4 Sensor-trust event evidence

The default west-facing Diamond façade detects the injected fault but remains open because its passive geometry already keeps the optimum at 0° at that time. The following targeted test therefore uses an **east-facing upright façade** to expose the control consequence.

Time (MYT)	Event	Measured (W/m²)	Expected (W/m²)	Trust decision	NeuroSkin angle	Naive angle	NeuroSkin lux	Naive lux	NeuroSkin load	Naive load
11:30	Injected dead pyranometer	0.0	816.12	Rejected	55°	0°	325.9	824.4	0.4509	0.7707
15:00	Genuine heavy cloud	8.0	281.63	Trusted	55°	0°	20.0	20.0	0.3778	0.5131

At 11:30, the near-zero sensor reading contradicted the solar almanac under a clear synthetic sky. NeuroSkin rejected the sensor and retained shading, while the naive controller opened. At 15:00, heavy cloud explained the low reading, so the sensor value was accepted.

A.5 Safety and movement-budget event evidence

Time (MYT)	Event	Mode	Target angle	Final angle	Measured irradiance (W/m ²)	Relative load	Result
14:00	Power loss	SAFE	60°	60°	721.02	0.3939	Optimiser bypassed; fail shaded
15:00	Marginal movement	HOLD	40°	60°	682.98	0.3887	Movement declined below threshold

The power-loss rule bypassed normal optimisation for four ten-minute ticks. After power returned, the optimiser requested 40°, but the predicted benefit did not clear the movement threshold, so the controller held 60°.

A.6 Façade geometry counterfactual

Geometry	pvlilb tilt	NeuroSkin daylight (%)	NeuroSkin load	NeuroSkin moves	Naive daylight (%)	Naive load	Naive moves
Diamond as built: 25° outward	115°	100.0	0.550	0	16.7	0.456	2
Upright counterfactual	90°	70.8	0.507	1	12.5	0.469	2

The Diamond Building's passive geometry reduced the need for adaptive movement and maintained daylight in the reference day. The upright counterfactual produced a lower relative load through additional active shading. The table should therefore be interpreted as an interaction between passive geometry and active control, not as a simple efficiency ranking.

Screenshots are from Jupyter Notebook output

A.7 Scope and metric definitions

Item	Definition
Simulation period	21 March 2026, 00:00–23:50 MYT
Resolution	144 ticks at ten-minute intervals
Environment	Seeded synthetic scattered-cloud day
Location	ST Diamond Building, Putrajaya
Daylight compliance	Percentage of occupied ticks within 300–700 lux where the controlled wall receives at least 200 W/m ² incident irradiance
Mean relative load	Arithmetic mean of the physics-inspired cooling-load index across occupied ticks
Movement count	Number of full-day final-angle changes
Sensor-fault ticks	Ticks where <code>sensor_trusted</code> is false
SAFE ticks	Ticks where controller mode is <code>SAFE</code>

The relative load includes a latent component that façade shading cannot eliminate. It must not be converted into HVAC energy or presented as kWh.

A.8 Simulation configuration

Parameter	Reference value	Targeted variation
Environment	Synthetic, scattered cloud	Unchanged
Date and timezone	21 March 2026, Asia/Kuala_Lumpur	Unchanged
Location	ST Diamond Building, Putrajaya	Unchanged
Seed	42	Seeds 0–9 for sensitivity check
Controlled façade	West	East for sensor-fault consequence
Façade geometry	115° pvlb tilt	90° upright counterfactual
Occupancy	1.0x	Unchanged
Wind	3 m/s override	Unchanged
Controller weights	Thermal 0.45; daylight 0.35; movement 0.15; risk 0.05	Unchanged
Candidate angles	0°–60° in 5° increments	Unchanged